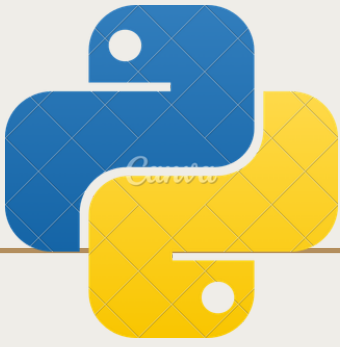
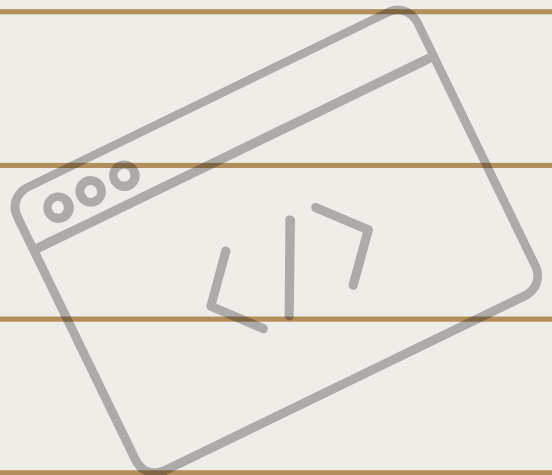




CONTENT



1. INTRODUCTION
2. PROBLEM STATEMENT
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INTRODUCTION

About the Project

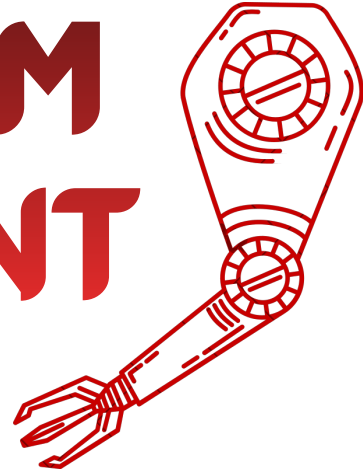


- Tic Tac Toe is a simple and well-known two-player game.
- The game is played on a 3×3 grid.
- One player uses the symbol X, while the other player uses the symbol O.
- Players take turns to place their symbols in empty spaces on the board.
- The main objective of the game is to place three identical symbols in a straight line.
- The straight line can be horizontal, vertical, or diagonal.
- This Tic Tac Toe game is developed using the Python programming language.
- The program accepts input from the players in the form of position numbers from 1 to 9.
- After every move, the updated game board is displayed.
- The program checks for a winning condition or a draw after each turn.

Coding Concept Behind the Game

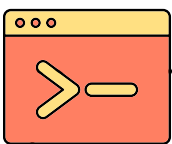
- **LISTS** - to store the game board.
- **FUNCTIONS** - to display the board and check results.
- **LOOPS** - to keep the game running.
- **if-else** - for decision making.
- **INPUT/OUTPUT** - for player interaction.

PROBLEM STATEMENT



Background and Need

- **Beginners often find it difficult to understand programming logic.**
- **A simple interactive program helps in improving logical thinking.**
- **Game-based programs make learning coding more interesting and practical.**
- **There is a need for a basic project to practice programming concepts.**



Objective and Scope



- **To develop a simple interactive program using Python.**
- **To accept user input and update the program accordingly.**
- **To apply decision-making logic to determine the program outcome.**
- **To use basic programming concepts such as lists, functions, loops, and conditions.**
- **The project is limited to logic implementation and does not include graphics.**



Creating an empty board using a list

```
1 board = [' '] * 9
```

Function to display the board

```
1 def display_board():
2     print()
3     print(" ", board[0], "|", board[1], "|", board[2])
4     print("----+---+----")
5     print(" ", board[3], "|", board[4], "|", board[5])
6     print("----+---+----")
7     print(" ", board[6], "|", board[7], "|", board[8])
8     print()
```

Function to check draw

```
1 def check_draw():
2
3     for i in board:
4         if i == " ":
5             return False
6
7     return True
```

Function to check win condition

```
1 def check_winner(player):
2
3     # Checking rows
4     if (board[0] == board[1] == board[2] == player or
5         board[3] == board[4] == board[5] == player or
6         board[6] == board[7] == board[8] == player):
7         return True
8
9     # Checking columns
10    if (board[0] == board[3] == board[6] == player or
11        board[1] == board[4] == board[7] == player or
12        board[2] == board[5] == board[8] == player):
13        return True
14
15    # Checking diagonals
16    if (board[0] == board[4] == board[8] == player or
17        board[2] == board[4] == board[6] == player):
18        return True
19
20    return False
21
```

Game instructions and starts the game

```
1 print("WELCOME TO TIC TAC TOE GAME")
2 print("Player 1 : X")
3 print("Player 2 : O")
4 display_board()
5
6 current_player = "X"
```

Looping after entering the value

```
1 while True:
2     print("Player", current_player, "turn")
3     position = int(input("Enter position (1-9): "))
```

Adjusting position to list index

```
1 position -= 1
```

Checking if position is empty

```
1 if board[position] == " ":
2     board[position] = current_player
3     display_board()
4
```

Checking for winner

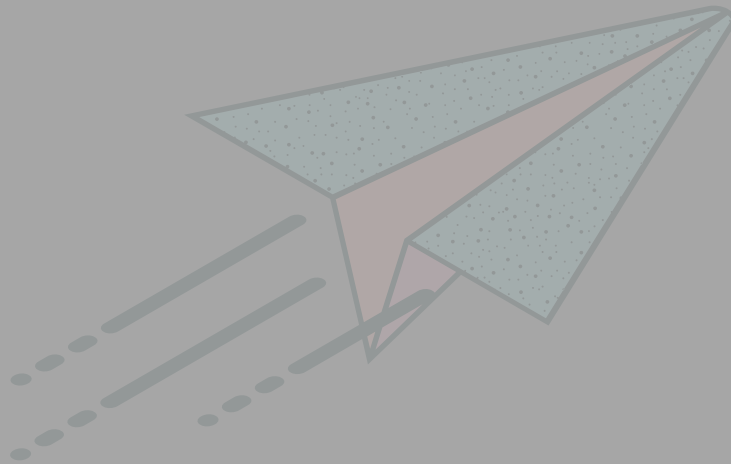
```
1 if check_winner(current_player):
2     print("Player", current_player, "wins the game!")
3     break
```

Checking for draw

```
1 if check_draw():  
2     print("The game is a draw!")  
3     break  
4
```

Switching player

```
1     if current_player == "X":  
2         current_player = "O"  
3     else:  
4         current_player = "X"  
5 else:  
6     print("Position already occupied. Try again.")
```



print(" final code")

```
1
2 # Creating an empty board using a list
3 board = [' '] * 9
4
5 # Function to display the board
6 def display_board():
7     print()
8     print(" ", board[0], "|", board[1], "|", board[2])
9     print("----+----+----")
10    print(" ", board[3], "|", board[4], "|", board[5])
11    print("----+----+----")
12    print(" ", board[6], "|", board[7], "|", board[8])
13    print()
14
15 # Function to check win condition
16 def check_winner(player):
17     # Checking rows
18     if (board[0] == board[1] == board[2] == player or
19         board[3] == board[4] == board[5] == player or
20         board[6] == board[7] == board[8] == player):
21         return True
22
23     # Checking columns
24     if (board[0] == board[3] == board[6] == player or
25         board[1] == board[4] == board[7] == player or
26         board[2] == board[5] == board[8] == player):
27         return True
28
29     # Checking diagonals
30     if (board[0] == board[4] == board[8] == player or
31         board[2] == board[4] == board[6] == player):
32         return True
33
```



```

35
36 # Function to check draw
37 def check_draw():
38     for i in board:
39         if i == " ":
40             return False
41     return True
42
43 # Main program
44 print("WELCOME TO TIC TAC TOE GAME")
45 print("Player 1 : X")
46 print("Player 2 : O")
47 display_board()
48
49 current_player = "X"
50
51 while True:
52     print("Player", current_player, "turn")
53     position = int(input("Enter position (1-9): "))
54
55     # Adjusting position to list index
56     position -= 1
57
58     # Checking if position is empty
59     if board[position] == " ":
60         board[position] = current_player
61         display_board()
62
63         # Checking for winner
64         if check_winner(current_player):
65             print("Player", current_player, "wins the game!")
66             break
67
68         # Checking for draw
69         if check_draw():
70             print("The game is a draw!")
71             break
72
73         # Switching player
74         if current_player == "X":
75             current_player = "O"
76         else:
77             current_player = "X"
78     else:
79         print("Position already occupied. Try again.")

```

Output :

After run the code

```
WELCOME TO TIC TAC TOE GAME
Player 1 : X
Player 2 : O

  |  |
--+--+
  |  |
--+--+
  |  |

Player X turn
Enter position (1-9): █
```

When X wins

```
  |  |
--+--+
  |  |
--+--+
  |  |

Player X turn
Enter position (1-9): 1

X |  |
--+--+
  |  |
--+--+
  |  |

Player O turn
Enter position (1-9): 3
```

```
  X |  | O
--+--+
  |  |
--+--+
  |  |
```

Player X turn
Enter position (1-9): 5

```
  X |  | O
--+--+
  | X |
--+--+
  |  |
```

Player O turn
Enter position (1-9): 2

```
  X | O | O
--+--+
  | X |
--+--+
  |  |
```

Player X turn
Enter position (1-9): 9

```
  X | O | O
--+--+
  | X |
--+--+
  |  | X
```

Player X wins the game!

When 0 wins

WELCOME TO TIC TAC TOE GAME

Player 1 : X

Player 2 : O

```

  |  |
  +--+
  |  |
  +--+
  |  |

```

Player X turn

Enter position (1-9): 1

```

X |  |
  +--+
  |  |
  +--+
  |  |

```

Player O turn

Enter position (1-9): 2

```

X | O |
  +--+
  |  |
  +--+
  |  |

```

Player X turn

Enter position (1-9): 3

```

X | O | X
  +--+
  |  |
  +--+
  |  |

```

Player O turn

Enter position (1-9): 4

```

X | O | X
  +--+
O |  |
  +--+
  |  |

```

Player X turn

Enter position (1-9): 6

```

X | O | X
  +--+
O |  | X
  +--+
  |  |

```

Player O turn

Enter position (1-9): 5

```

X | O | X
  +--+
O | O | X
  +--+
  |  |

```

Player X turn

Enter position (1-9): 7

```

X | O | X
  +--+
O | O | X
  +--+
X |  |

```

Player O turn

Enter position (1-9): 8

```

X | O | X
  +--+
O | O | X
  +--+
X | O |

```

Player O wins the game!

When draw

WELCOME TO TIC TAC TOE GAME

Player 1 : X

Player 2 : O

----	----
----	----

Player X turn

Enter position (1-9): 1

X		
----	----	----
----	----	----

Player O turn

Enter position (1-9): 3

X		O
----	----	----
----	----	----

Player X turn

Enter position (1-9): 2

X	X	O
----	----	----
----	----	----

Player O turn

Enter position (1-9): 4

X	X	O
----	----	----
O		
----	----	----

Player X turn

Enter position (1-9): 6

X		X		O
----	----	----	----	----
O			X	
----	----	----	----	----

Player O turn

Enter position (1-9): 5

X		X		O
----	----	----	----	----
O		O		X
----	----	----	----	----

Player X turn

Enter position (1-9): 9

X		X		O
----	----	----	----	----
O		O		X
----	----	----	----	----
		X		

Player O turn

Enter position (1-9): 8

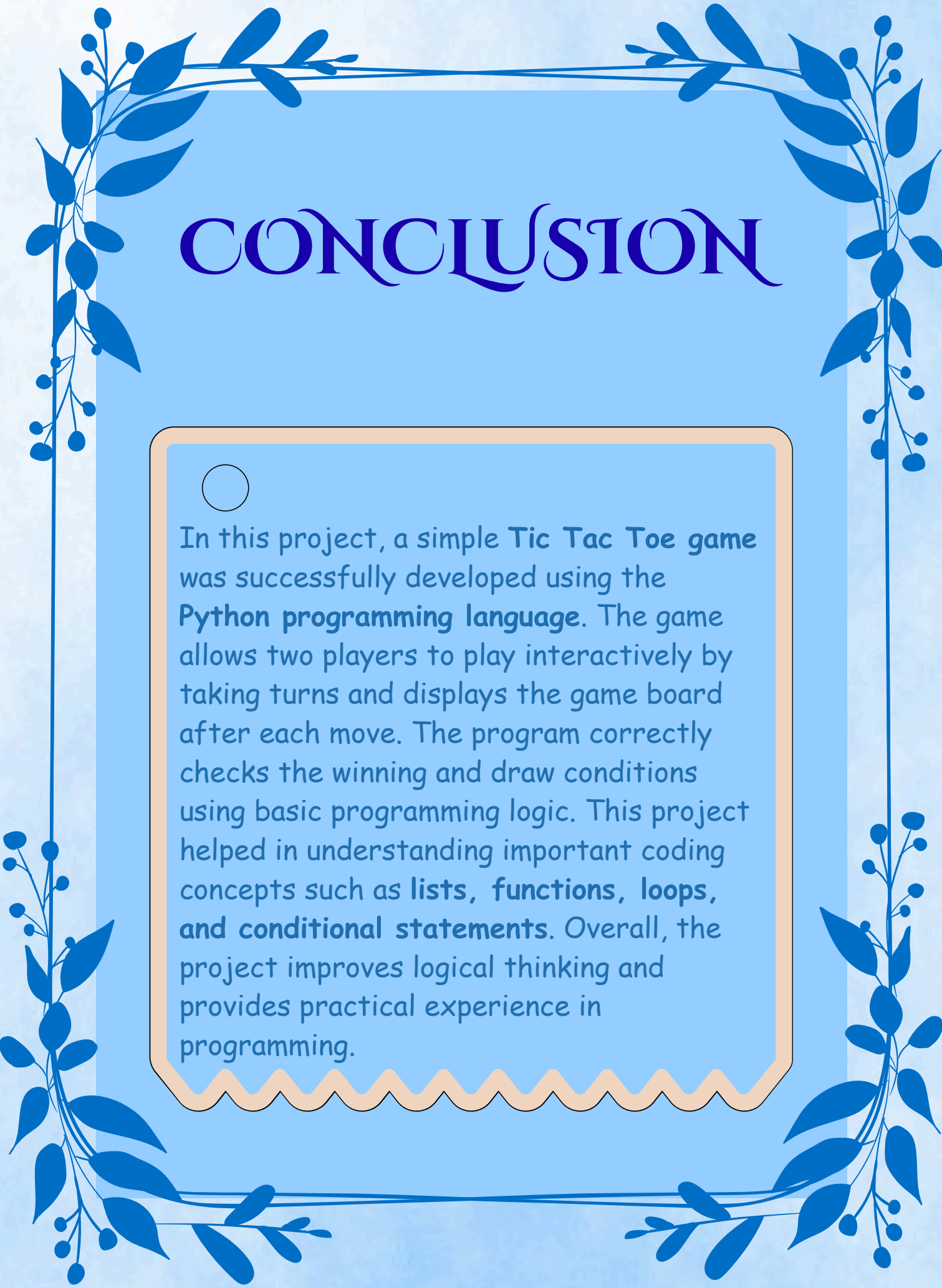
X		X		O
----	----	----	----	----
O		O		X
----	----	----	----	----
	O		X	

Player X turn

Enter position (1-9): 7

X		X		O
----	----	----	----	----
O		O		X
----	----	----	----	----
X		O		X

The game is a draw!



CONCLUSION



In this project, a simple Tic Tac Toe game was successfully developed using the Python programming language. The game allows two players to play interactively by taking turns and displays the game board after each move. The program correctly checks the winning and draw conditions using basic programming logic. This project helped in understanding important coding concepts such as **lists, functions, loops, and conditional statements**. Overall, the project improves logical thinking and provides practical experience in programming.



Bibliography

**Python Software Foundation,
Python Official Documentation.**

**Class notes and study materials
provided by the teacher.**

**Online tutorials and reference
materials on Python programming.**

